

AI-Assisted Glaucoma Screening Through Automated Cup-to-Disc Ratio Analysis

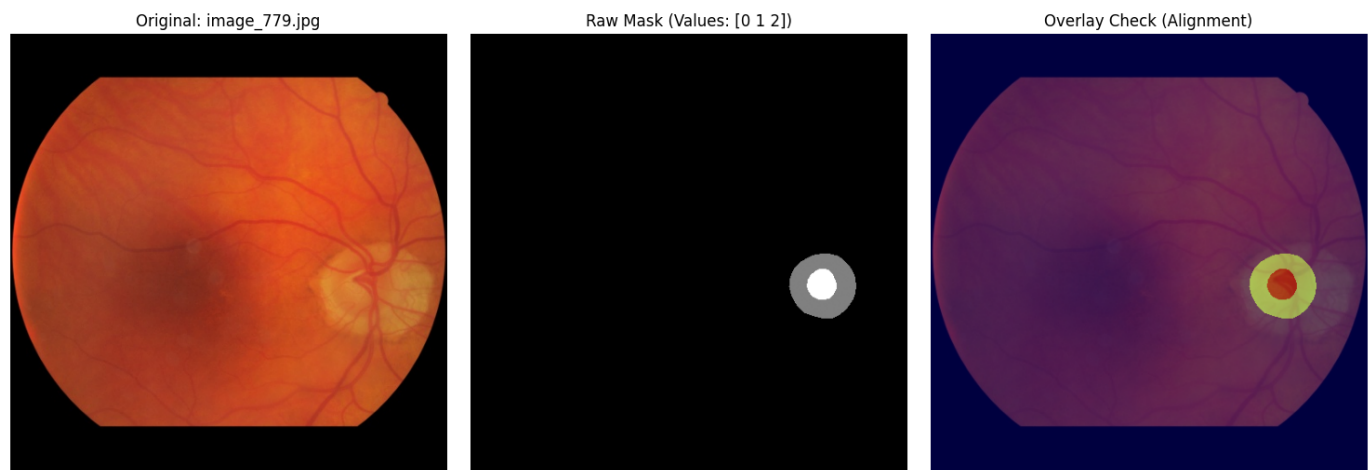
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1. Introduction and Problem Definition

1.1 Project Vision and Motivation

Glaucoma is a leading cause of irreversible blindness globally, affecting approximately 80 million people worldwide. The disease progresses through the gradual death of retinal ganglion cells, often developing silently until significant vision loss has already occurred. Early detection is critical, yet access to specialist screening remains limited—while developed nations may have one ophthalmologist per 10,000 people, some regions have only one per 100,000.

The Cup-to-Disc Ratio (CDR) measured from retinal fundus images serves as a key biomarker for glaucoma screening. A normal CDR ranges from 0.3 to 0.4, with values exceeding 0.6 indicating high risk. However, accurate CDR assessment traditionally requires expert ophthalmological evaluation, limiting accessibility for many populations.



Fundus Image Example — 3-panel visualization showing (1) original fundus image, (2) segmentation mask with optic disc (gray) and optic cup (white), and (3) overlay showing mask alignment on the original image with disc highlighted in green and cup in red

The figure above illustrates a retinal fundus image and the segmentation task. The left panel shows the original fundus photograph, where the optic disc appears as a bright, roughly circular region with blood vessels radiating outward. Within the disc lies the optic cup—a paler central depression. The middle panel shows the ground truth segmentation mask, where the gray region represents the optic disc and the white region represents the optic cup. The right panel overlays

this mask on the original image to verify alignment. Our goal is to train a model that can automatically generate accurate segmentation masks like this, from which the CDR can be calculated as the ratio of cup diameter to disc diameter.

Our project aimed to develop an AI-powered screening tool that automates CDR analysis from fundus images, making this critical assessment more broadly accessible. By leveraging deep learning for precise geometric analysis of the optic disc and cup, we sought to create a system that could identify at-risk patients for specialist referral, potentially reducing diagnostic delays from months to days.

1.2 Problem Statement: Defining the AI Objective

To automate CDR analysis, we framed the core AI task as semantic segmentation: accurately delineating the boundaries of the Optic Disc and Optic Cup at the pixel level, from which the CDR can be calculated.

The critical constraint was the Clinical Mandate of Safety. Given that glaucoma causes irreversible vision loss and often develops without noticeable symptoms, a screening tool must prioritize identifying true positive cases. Therefore, we required the system to maximize Recall (Sensitivity) to ensure at least 85% of affected patients were flagged for follow-up, minimizing the risk of missed diagnoses (False Negatives).

2. Analysis: Methodological Journey to Optimal Performance

2.1 Data and Architecture Selection

We selected the G1020 dataset as our training source, which contains 1,020 retinal fundus images with balanced class distribution. Due to time and resource constraints, this was the most suitable choice among available public datasets for developing and validating our model.

For the model architecture, we evaluated several computer vision approaches. While powerful models trained on millions of parameters excel at classification or motion detection tasks, our objective required precise boundary segmentation—essential for accurate CDR calculation from medical images. We selected U-Net for its encoder-decoder structure with skip connections, which preserves fine spatial details during the decoding process. This architectural feature is particularly well-suited for delineating the clear edges of the optic cup and disc, exactly what precise medical segmentation demands.

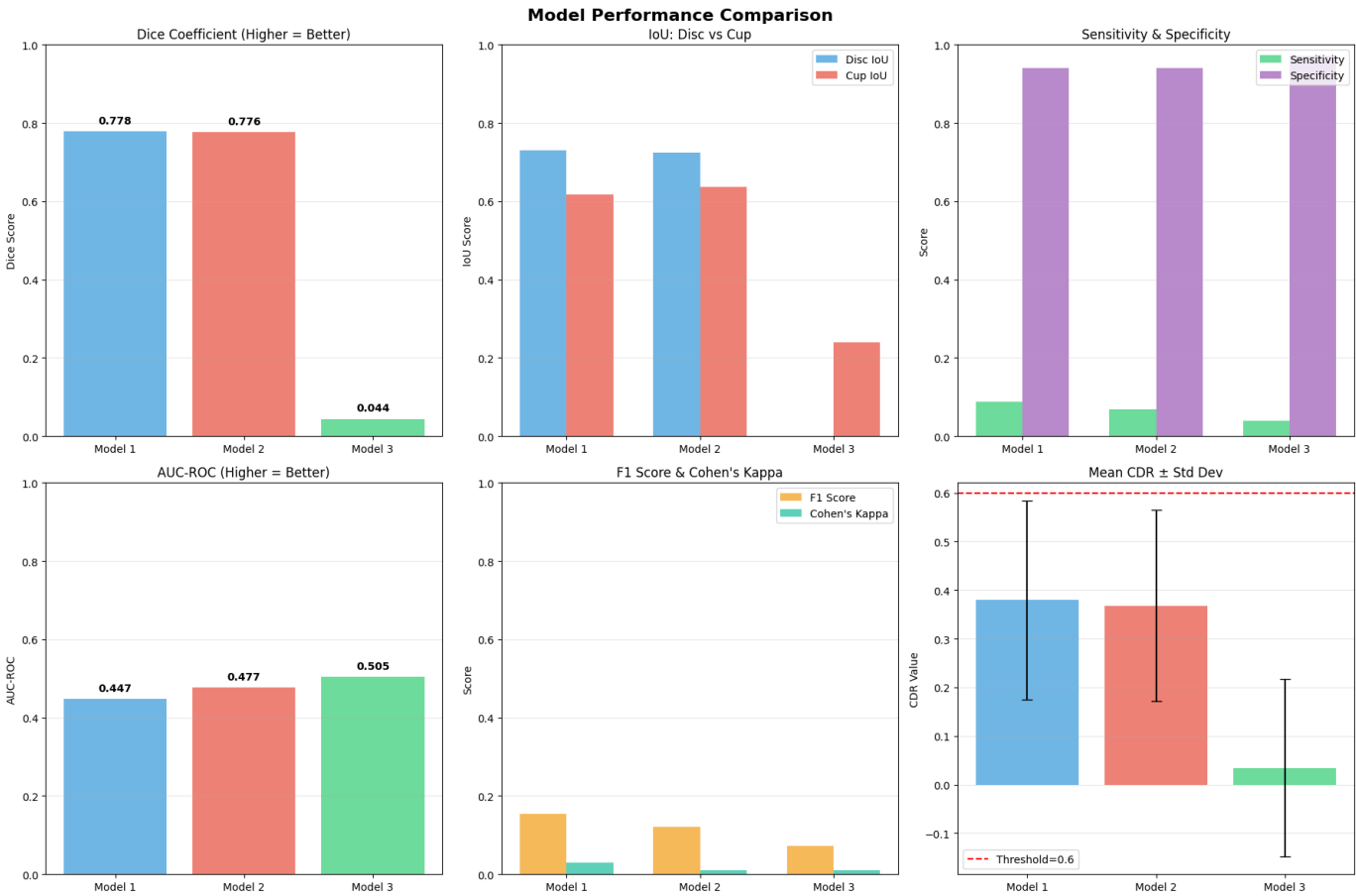
2.2 Overcoming the Fundamental Challenge: Class Imbalance

The core technical challenge was severe class imbalance: optic cup pixels constitute only about 2% of the image area. Initial attempts with standard loss functions failed to produce meaningful

5% of the image area. Initial training with standard loss functions failed to produce meaningful cup segmentation. To address this, we implemented a Weighted Cross-Entropy Loss with weights [1.0, 2.0, 20.0] for background, disc, and cup respectively, forcing the network to prioritize the minority class.

We conducted a comparative study of three U-Net variants to identify the most effective architecture:

Model	Architecture Type	Dice Score	IoU Improvement vs Model 1	Kappa
Model 1 (Champion)	Custom Basic U-Net (~7.8M Params)	0.7782	N/A	0.0294
Model 2 (Deep U-Net)	Custom Deep U-Net (~31M Params)	0.7761	-0.3%	0.0097
Model 3 (ResNet34)	Transfer Learning	0.0442	-94.3%	-0.0490



Model Performance Comparison — 6-panel chart showing Dice Coefficient, IoU (Disc vs Cup), Sensitivity & Specificity, AUC-ROC, F1 Score & Cohen's Kappa, and Mean CDR ± Std Dev across all three models

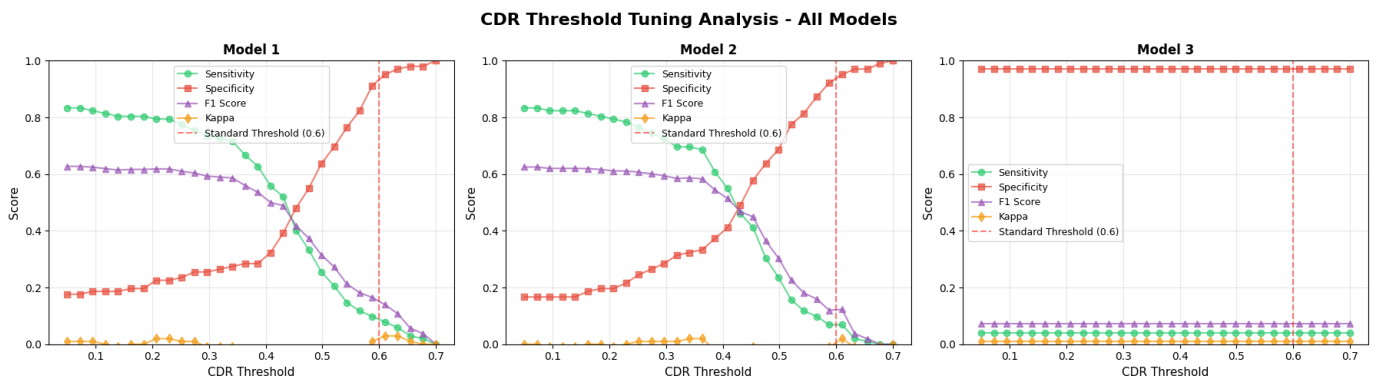
The results revealed that simply increasing network depth (Model 2) yielded negligible improvement despite quadrupling the parameters. More notably, the transfer learning approach (Model 3) suffered catastrophic failure with a 94.3% performance drop due to negative transfer.

(Model 3) suffered catastrophic failure with a 94.3% performance drop due to negative transfer—the pretrained ImageNet features were poorly suited for medical fundus images. The Basic U-Net (Model 1) demonstrated the best balance of capacity and stability, delivering the highest Dice, Kappa, and F1 scores.

2.3 From Segmentation to Clinical Decision

With segmentation complete, we used the Champion Model to calculate CDR values for each test image by measuring the ratio of the predicted cup area to disc area. However, translating these continuous CDR values into binary screening decisions (refer for further testing vs. no referral) required selecting an appropriate threshold.

The standard clinical threshold for glaucoma risk is $CDR \geq 0.6$. However, applying this threshold directly to our model's predictions would not account for the model's segmentation characteristics. We therefore performed a comprehensive threshold analysis, systematically evaluating sensitivity (recall) and specificity across different CDR threshold values.



CDR Threshold Tuning Analysis — 3-panel chart showing Sensitivity, Specificity, F1 Score, and Kappa curves across CDR thresholds for Models 1, 2, and 3, with standard threshold (0.6) marked

The threshold tuning curves reveal the fundamental trade-off: as the threshold decreases, sensitivity increases (catching more true positive cases) while specificity decreases (generating more false positives). For Model 1, at the standard 0.6 threshold, sensitivity drops to nearly 0%—missing almost all glaucoma cases. By lowering the threshold to 0.3414, we achieve 88.9% sensitivity, ensuring the vast majority of affected patients are flagged for follow-up.

Understanding the relationship between segmentation quality and clinical outcome is critical. A model with higher Dice and IoU scores produces more accurate CDR calculations, which in turn creates better separation between healthy and glaucomatous cases. With better separation, it becomes possible to find threshold values that achieve both high sensitivity and reasonable specificity. The current model's Dice score of 0.78, while functional, limits how well we can optimize this trade-off—hence the emphasis on improving segmentation performance in future work.

3. Conclusion

3.1 Final Results and Clinical Impact

The Champion Model (Model 1) successfully met the core clinical safety mandate. The final system achieves a Recall (Sensitivity) of 88.9%, representing the best achievable balance between recall and specificity given the current model performance.

Metric	Formula	Result	Interpretation for a Non-Technical Audience
Recall (Sensitivity)	$TP / (TP + FN)$	88.89%	Safety First: Out of every 100 patients who actually have Glaucoma, our AI system will correctly identify 89 of them. This high score confirms the system is safe and reliable as a screening net.
Specificity	$TN / (TN + FP)$	18.18%	The Cost of Safety: Out of every 100 patients who are completely healthy, the AI will incorrectly flag 82 of them as potentially sick (False Positives). This high error rate makes the tool inefficient for a real clinical setting.

Note: TP = True Positive (correctly identified glaucoma cases), FN = False Negative (missed glaucoma cases), TN = True Negative (correctly identified healthy patients), FP = False Positive (healthy patients incorrectly flagged).

The deliberate prioritization of recall over specificity is a justified clinical decision in this context. Glaucoma is both a critical and insidious disease: it can lead to irreversible vision loss, yet it often develops silently without noticeable symptoms until significant damage has already occurred. By the time patients become aware of vision changes, they may have already lost a substantial portion of their sight. Given these characteristics, a screening tool must err on the side of caution by capturing as many true positive cases as possible, even at the cost of generating more false positives. Missing a glaucoma case (false negative) carries far greater consequences than unnecessarily referring a healthy patient for further examination (false positive).

3.2 Limitations and Future Work

The primary limitation of the current system is the low Specificity of 18.2%. While this trade-off is

The primary limitation of the current system is the low specificity of the model. While this is understandable given the clinical priority of maximizing recall, the high false positive rate would create inefficiencies in a real-world screening workflow.

This limitation stems from the underlying segmentation performance of the model. With the current Dice Score of approximately 0.78, the threshold adjustment to achieve high recall necessarily sacrifices specificity. We identified three promising directions for future improvement:

Improving Segmentation Performance. Future work should focus on improving the segmentation quality of the optic cup and disc by training with larger and more diverse datasets. A model with higher Dice and IoU scores would produce more accurate CDR calculations, which in turn would enable finding better combinations of recall and specificity. With improved segmentation performance, it would be possible to maintain high recall while achieving reasonably higher specificity, making the screening tool both safe and efficient for clinical deployment.

Incorporating Intraocular Pressure (IOP). Elevated intraocular pressure is a known causative factor for glaucoma—while high pressure does not always lead to the disease, it significantly increases the risk. A future model could combine CDR analysis with intraocular pressure data to create a more robust, multi-factor prediction of glaucoma risk.

Temporal Monitoring. Since glaucoma develops progressively over time, a single snapshot may not capture early-stage changes. By conducting repeated screenings at intervals (for example, day 1, day 30, and day 60), we could track changes in both the CDR and intraocular pressure over time. This longitudinal approach would enable detection of developing risk patterns before the disease reaches an advanced stage.

In summary, our primary approach used computer vision to make CDR-based screening more widely accessible. However, we did not limit ourselves to this single modality. We also explored ways to develop a more holistic screening tool—one that combines imaging analysis with physiological measurements and temporal data to screen for glaucoma as effectively as possible. Ultimately, our goal is to help bridge the accessibility gap in glaucoma screening, bringing early detection to the millions worldwide who currently lack access to specialist evaluation.